

HW2

RJ Cass

```
suppressPackageStartupMessages(library(ggplot2))
suppressPackageStartupMessages(library(patchwork))

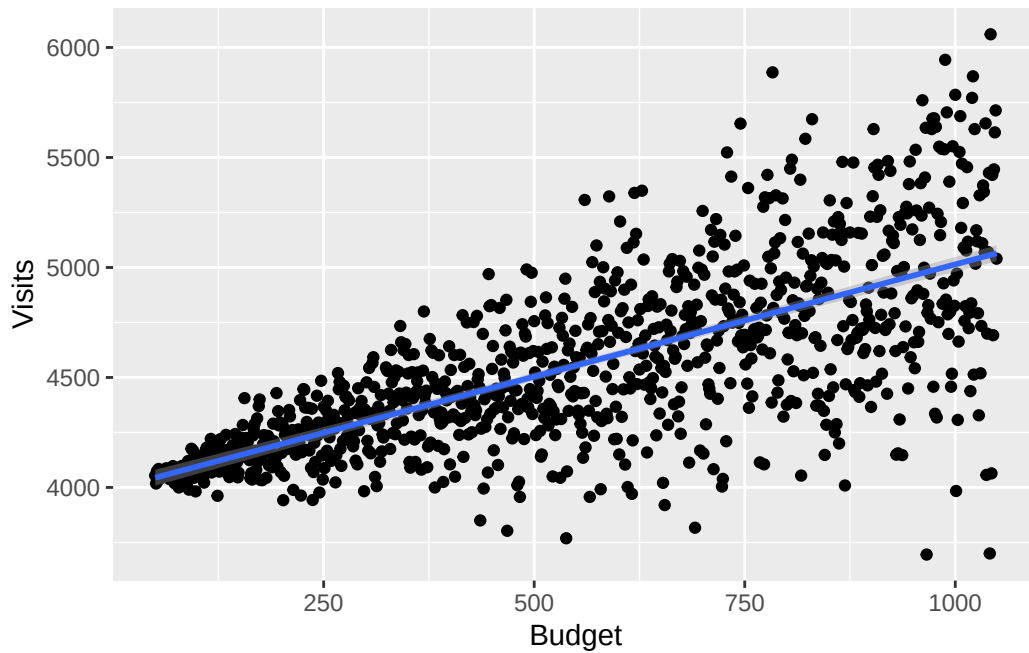
dt <- read.csv('website.csv')
```

1

(a)

```
p11 <- ggplot(dt, aes(x = Budget, y = Visits)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE)
p11
```

`geom\_smooth()` using formula = 'y ~ x'



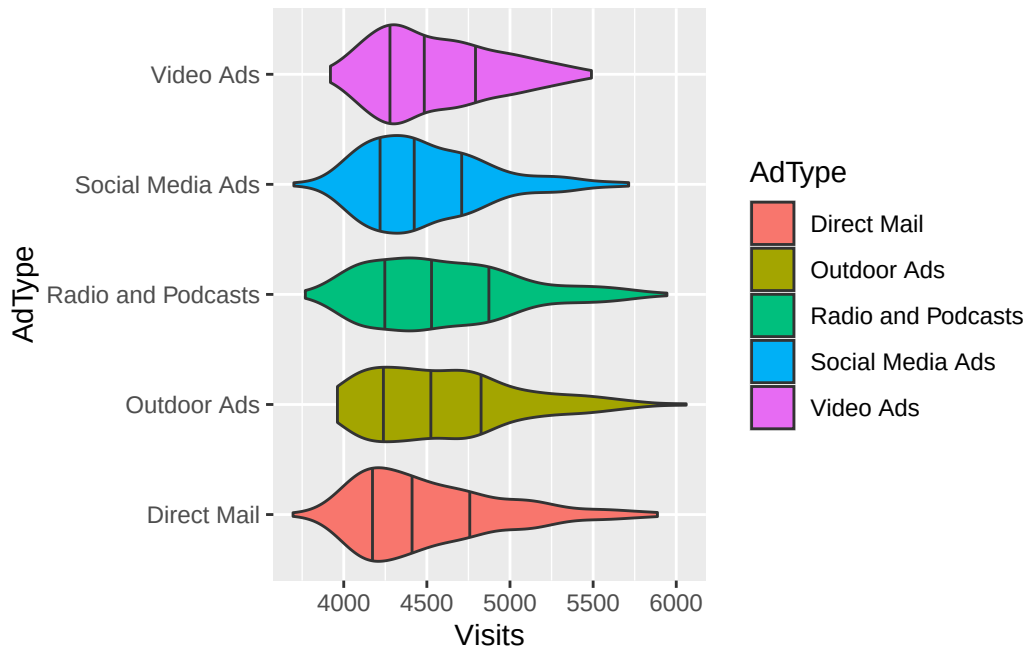
(b)

```
p12 <- ggplot(dt, aes(x = Visits, y = AdType, fill = AdType)) +  
  geom_violin(draw_quantiles=c(0.25, .5, .75))
```

Warning: The `draw\_quantiles` argument of `geom\_violin()` is deprecated as of ggplot2 4.0.0.

i Please use the `quantiles.linetype` argument instead.

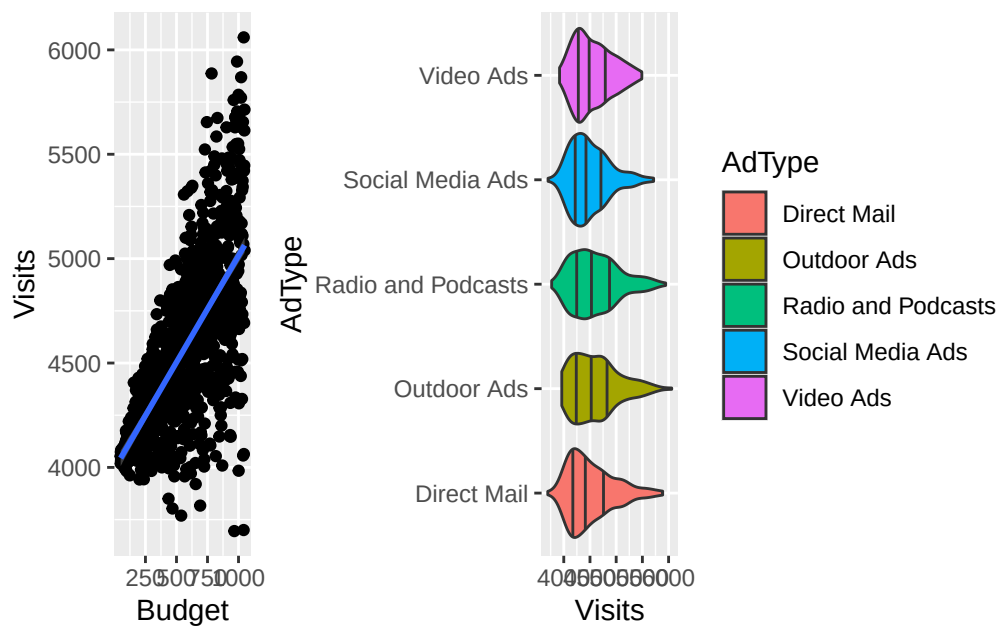
```
p12
```



Plotted together

p11 + p12

``geom_smooth()`` using formula = 'y ~ x'



(c)

```
mod_budget <- lm(Visits ~ Budget, data = dt)
mod_adtype <- lm(Visits ~ AdType, data = dt)
mod_full <- lm(Visits ~ Budget + AdType, data = dt)

summary(mod_full)
```

Call:

```
lm(formula = Visits ~ Budget + AdType, data = dt)
```

Residuals:

Min	1Q	Median	3Q	Max
-1313.54	-143.98	3.61	140.65	1095.48

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3995.4368	26.0962	153.104	<2e-16 ***
Budget	1.0167	0.0322	31.571	<2e-16 ***
AdTypeOutdoor Ads	17.6226	28.9569	0.609	0.543
AdTypeRadio and Podcasts	31.7844	29.0028	1.096	0.273
AdTypeSocial Media Ads	-40.2877	29.3655	-1.372	0.170
AdTypeVideo Ads	-10.3677	28.7366	-0.361	0.718

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 293 on 994 degrees of freedom

Multiple R-squared: 0.506, Adjusted R-squared: 0.5035

F-statistic: 203.6 on 5 and 994 DF, p-value: < 2.2e-16

```
anova(mod_budget, mod_full)
```

Analysis of Variance Table

Model 1: Visits ~ Budget

Model 2: Visits ~ Budget + AdType

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	998	85928668				
2	994	85343805	4	584863	1.703	0.1471

```
anova(mod_adtype, mod_full)
```

Analysis of Variance Table

Model 1: Visits ~ AdType

Model 2: Visits ~ Budget + AdType

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	995	170919701				
2	994	85343805	1	85575896	996.7	< 2.2e-16 ***

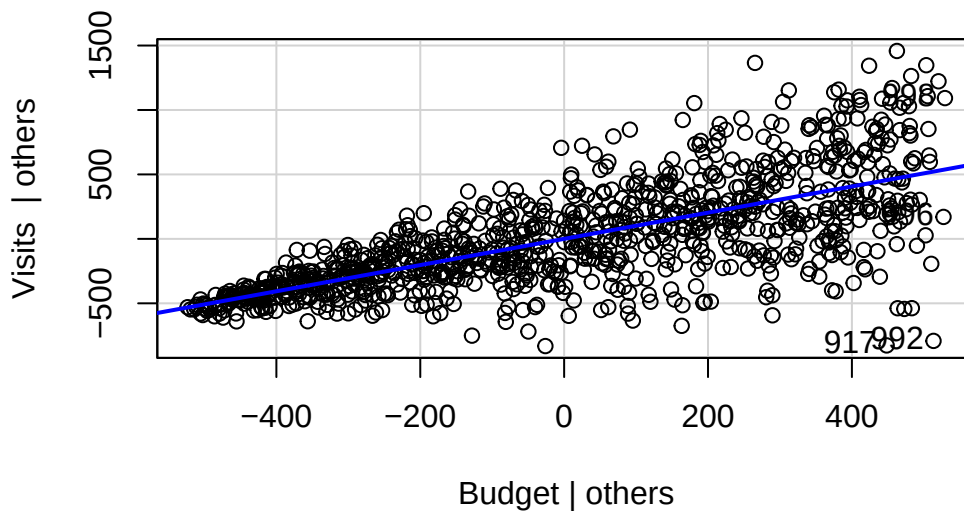
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

AdType does appear to be significant. Budget does not appear to be significant.

(d)

Linearity

```
car::avPlots(mod_full, terms = ~Budget)
```

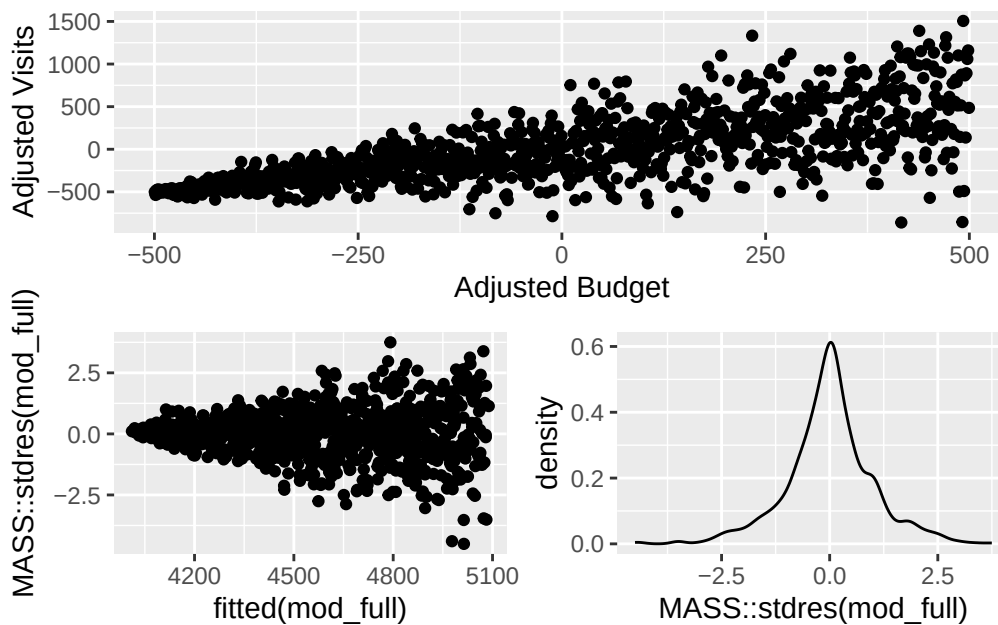


```

p141 <- ggplot(dt, aes(x = (Budget - mean(Budget)), y = (Visits - mean(Visits)))) +
  geom_point() +
  labs(x = "Adjusted Budget", y = "Adjusted Visits")
p142 <- ggplot(dt, aes(x = fitted(mod_full), y = MASS::stdres(mod_full))) +
  geom_point()
p143 <- ggplot(dt, aes(x = MASS::stdres(mod_full))) +
  geom_density()

p141 / (p142 + p143)

```



(e)

```

p151 <- ggplot(dt, aes(x = log(Budget), y = Visits)) +
  geom_point() +
  geom_smooth()
p152 <- ggplot(dt, aes(x = Budget, y = log(Visits))) +
  geom_point() +
  geom_smooth()
p153 <- ggplot(dt, aes(x = log(Budget), y = log(Visits))) +
  geom_point() +
  geom_smooth()
p154 <- ggplot(dt, aes(x = sqrt(Budget), y = log(Visits))) +

```

```
geom_point() +  
geom_smooth()
```

```
(p151 + p152) / (p153 + p154)
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'  
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```

